MTH313M ASSIGNMENT 2

Instructions:

(i) Each question is of 10 marks.

(ii) Only one question will be evaluated. The question number (to be evaluated) will be declared after the deadline for submission.

(iii) Deadline for submission: **6:00 pm IST, Friday April 24, 2026**.

(iv) **Assignment submission will be through the HelloIITK portal**. Email submissions and any late submission through the HelloIITK portal will NOT be graded.

(v) Please **upload one single PDF file** containing all your answers.

(vi) Provide all details/justifications.

Question 1. (a) (5 marks) Consider a pure Death process $\{X_t\}_{t\geq 0}$ with initial population i and the infinitesimal death rates as $\mu_n = n\mu, n = 1, 2, \cdots$, for some $\mu > 0$. Find $\mathbb{P}(X_t = n)$, $\mathbb{E}X_t$ and $Var(X_t)$.

(b) (5 marks) Consider an infinitely many server queueing problem with an exponential service time distribution with parameter $\mu > 0$. Suppose customers arrive in batches with the interarrival times following an exponential distribution with parameter $\lambda > 0$. The number of arrivals in a batch follows the Geometric distribution with parameter $p \in (0, 1)$, i.e. batch size k happens with probability $p^{k-1}(1-p), k = 1, 2, \cdots$. Show that the queue size is a continuous time Markov chain and find the infinitesimal matrix A appearing in the Kolmogorov's Forward/Backward Equations.

Question 2. (a) (5 marks) Let $\{X_t\}_{t\geq 0}$ be a pure Birth process. Derive the p.m.f. of X_t for any $t \geq 0$ from the first principles.

(b) (5 marks) Consider a Yule process $\{X_t\}_{t\geq 0}$ with initial population 1 and infinitesimal birth rate $\lambda > 0$. Given $x \geq 0$, define $N_t^{(x)}$ as the number of members of the population at time t of age less or equal to x. Find the distribution function of $N_t^{(x)}$.